

Zhengyang Yu

Curriculum Vitae

San Diego & Bay Area

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA

Expected Graduation: June 2028

Relevant Coursework: Systems Programming, Computer Organization, Advanced Data Structures, Design and Analysis of Algorithms, Machine Learning

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium**, *Personal Project*

- Designed and built C++ firmware for an ESP32-S3 device integrating touch and IMU peripherals, authenticated HTTPS access, multi-account support, and a responsive interactive interface.
- Implemented concurrent FreeRTOS tasks, adaptive and rate-limit-aware polling, and RGB565 frame-buffer rendering in PSRAM; added bounded persistent logging and runtime diagnostics to troubleshoot firmware behavior and improve reliability.

2025 **HTML to ePub Converter**, *Personal Project*

- Engineered a Python content-processing pipeline with modular HTML parsing, metadata embedding, and structured error handling for reliable ePub generation.

2024 **Home Assistant Presence Plugin**, *Personal Project*

- Built a Python integration that converts entry and exit sensor events into real-time room-occupancy state.
- Designed an event-driven REST API workflow with explicit state transitions for reliable smart-lighting automation and easier testing and debugging.

Skills

Programming C++, C, Python, Java

Systems ESP32-S3, FreeRTOS, Linux, firmware development, hardware-software integration, concurrent and event-driven systems

Engineering Data Structures, Algorithms, Object-Oriented Design, Software Testing and Validation, Performance Optimization, Debugging

Tools Git, Docker, PlatformIO, VS Code, HTTP/HTTPS, REST APIs, structured logging and runtime diagnostics

AI Tooling Codex-assisted implementation, refactoring, debugging, code validation, and technical documentation

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch

Diagnosed computer and software issues and translated technical solutions into clear guidance for community members.

2025–Present Member, UCSD Envision Makerspace

Prototype embedded and electronics projects using makerspace fabrication tools.